ECE 447 Data Analysis and Machine Learning for Engineers

Winter 2026 - January 05 to April 15

Class time: Tuesday, Thursday 11:00-12:20 Location: CSC B-10

Instructor:

Bahareh Tolooshams, PhD

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Donadeo Innovation Centre For Engineering 5-289
Office Hours: Tuesday 3 PM - 4 PM, Wednesday 9:15 AM - 10:30 AM

Course Description:

*3.5 (fi) (either term, 3-1S-0) The course introduces basic concepts and techniques of data analysis and machine learning. Topics include: data preprocessing techniques, decision trees, nearest neighbor algorithms, linear and logistic regressions, clustering, dimensionality reduction, model evaluation, deployment methods, and emerging topics.

Prerequisites: ECE 220 or CMPUT 275, and ECE 342 or STAT 235, or consent of instructor

Course synchronous and asynchronous content delivery schedule:

The sections of ECE 447 will cover relatively similar concepts; the delivery would be distinct and asynchronous.

Land Acknowledgment:

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

TA Information:

Jaeho Chung, jaeho6@ualberta.ca

Seminar Sections:

Section	Day	Time	Location
SEM - J32	Wednesday	8:00AM to 8:50AM	CCIS 1-160

Course Objectives & General Content:

The course is an introduction to Machine Learning. The goals are:
- goal 1: to teach you the basics of Data Analysis
- goal 2: to expose you to algorithms and techniques of Machine Learning (ML)
- goal 3: to teach you to mathematical concepts behind the machine learning methods helpful in working with data analysis/processing methods

- goal 4: to give you the background required for using Machine Learning Operationalization Management (MLOps)

Learning Outcomes:

By the end of this course, students should be able to:

1. At the end of the course, the students are expected to learn principles of Data Analysis that include using the most common algorithms and techniques of Machine Learning:
   - Learn the basics of processes suitable for data analysis
   - Recognize different approaches for building data models
   - Understand and apply supervised methods for building prediction models
   - Understand and use strategies for the unsupervised processing of data and analysis of the obtained results
   - Learn a selected set of mathematical concepts needed to understand techniques for data modeling
   - Understand the Life Cycle of developed data models

2. Students will be able to:
   - Construct data/prediction models (regression, decision trees, ensemble models, SVM, basic NN)
   - Conduct analysis of models' performance (different evaluation approaches)
   - Apply unsupervised learning methods, and dimensionality reduction and anomaly detection
   - Identify various data distributions, able to perform vector and matrix operations, gradient descent
   - Reason about stages related to model development, its deployment, and management

Marking Scheme:

Activity	(A)Synchronous	Due/Scheduled	Weight
Assignment 1	Asynchronous		5%
Assignment 2	Asynchronous		5%
Assignment 3	Asynchronous		5%
Assignment 4	Asynchronous		5%
Project	Asynchronous	20 April 2026	25%
Midterm	Synchronous	26 February 2026	25%
Final	Synchronous	21 April 2026	30%

The Faculty recommended grade point average for a 400 level course is 3.1.Instructors have the leeway to deviate from this average and can assign grades based on their own scheme. All grades are approved by the department chair (or delegate). The office of the Dean has final oversight on all grades.

Term Work

All term work solutions will be posted no later than the last day of classes. All term work will be returned to students by the final day of classes, with the exception of major term work due in the last week of classes. The latter will be returned by the day of the final examination or the last day of the examination period if there is no final examination in the course as per university policy; instructors will make accommodations to return these term work. It is the responsibility of the student to pick up all their term work at the specified

time and place. Any unreturned term work, shall be retained and then shredded six months after the deadline for reappraisal and grade appeals. Final examinations will be kept for one year as required by university guidelines and the Government of Alberta's Freedom of Information and Protection of Privacy Act.

Calculator Policy

There is no calculator policy in this course; students are free to use the calculator they wish for all assessments.

Expectations for AI use

In this course, we commit to AI use guided by ethical and transparent principles. While students are allowed to use advanced automated tools (such as ChatGPT or Dall-E 2) for certain written assignments, it is crucial to adhere to the following guidelines:
Seek prior approval from the instructor for AI use in specific assignments.
When allowed, clearly attribute and cite any AI-generated content in your work, including prompts and AI outputs as part of your academic record. Include an additional reflection component in your assessments, discussing how AI tools contributed to your learning process.
IMPORTANT: Please note that AI use is strictly prohibited in assessments and assignments not approved by the instructor. Failure to abide by this guideline may be considered an act of cheating and a violation as outlined in the relevant sections of University of Alberta (November 2022) Code of Student Behaviour .

Text and References (Recommended):

John D. Kelleher, Brian Mac Namee and Aoife D'Arcy,
Fundamentals of Machine Learning for Predictive Data Analytics
MIT Press; 2nd Edition (October 20, 2020).

Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong,
Mathematics for Machine Learning
Cambridge University Press (April 23 2020).

Aurélien Géron,
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems
O'Reilly Media; 2nd Edition (September 5, 2019).

Emmanuel Raj,
Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale
Packt Publishing (April 19, 2021)

Website:

Did you know that the University of Alberta has various low-to-no-cost services to help students succeed? Visit http://www.deanofstudents.ualberta.ca/ for information about the academic, wellness, and various other support services available to U of A students. It's never too early or too late to seek help!

University and faculty policies

Respect and professionalism

The Faculty of Engineering is committed to fostering and protecting an equitable, inclusive, and respectful work and study environment in line with University of Alberta policies and professional engineering industry standards.

The faculty prepares students to uphold industry standards to become a Professional Engineer (P.Eng). Therefore, respect, professionalism, and accountability must be upheld within the Faculty of Engineering and the University of Alberta.

Academic integrity and student conduct

The University of Alberta is committed to the highest standards of academic integrity and honesty, as well as maintaining a learning environment that fosters the safety, security, and the inherent dignity of each member of the community, ensuring students conduct themselves accordingly. Students are expected to be familiar with the standards of academic honesty and appropriate student conduct, and to uphold the policies of the University in this respect.

Students are particularly urged to familiarize themselves with the provisions of the Student Academic Integrity Policy and the Student Conduct Policy, and avoid any behaviour that could

potentially result in suspicions of academic misconduct (e.g., cheating, plagiarism, misrepresentation of facts, participation in an offence) and non-academic misconduct (e.g., discrimination, harassment, physical assault). Academic and non-academic misconduct are taken very seriously and can result in suspension or expulsion from the University.

All students are expected to consult the Academic Integrity website for clarification on the various academic offences. All forms of academic dishonesty are unacceptable at the University. Unfamiliarity of the rules, procrastination or personal pressures are not acceptable excuses for committing an offence. Listen to your instructor, be a good person, ask for help when you need it, and do your own work -- this will lead you toward a path to success. Any academic integrity concern in this course will be reported to the College of Natural and Applied Sciences. Suspected cases of non-academic misconduct will be reported to the Dean of Students. The College, the Faculty, and the Dean of Students are committed to student rights and responsibilities, and adhere to due process and administrative fairness, as outlined in the Student Academic Integrity Policy and the Student Conduct Policy. Please refer to the policy websites for details on inappropriate behaviours and possible sanctions.

The College of Natural and Applied Sciences (CNAS) has created an Academic Integrity for CNAS Students eClass site. Students can self-enroll and review the various resources provided, including the importance of academic integrity, examples of academic misconduct & possible sanctions, and the academic misconduct & appeal process. Students can also complete assessments to test their knowledge and earn a completion certificate.

"Integrity is doing the right thing, even when no one is watching." -- C.S. Lewis

The Faculty of Engineering expects an environment free of harassment, discrimination, and bullying. We encourage you to talk to the Office of Safe Disclosure and Human Rights about experiences, questions, or concerns. Additional resources and support for students are attached below.

Engineering students studying in the province of Alberta must also follow the Code of Ethics set by the Association of Professional Engineers and Geoscientists of Alberta (APEGA).

Course outline policies, course requirements, evaluation and grading information can be found in the University Calendar.

Safety during learning activities

In all Faculty of Engineering courses, labs, seminars or other learning activities, safety is of paramount importance. In some cases, laboratory work in a program requires high standards for risk management to keep potential hazards safely under control.

Anyone found to be unable to function safely in the class, lab, seminar or other learning activity may be asked to leave or be removed for their and the safety of other participants and instructors in alignment with the Student Academic Integrity Policy and Student Conduct Policy. As members, or prospective members, of the engineering profession, it is your responsibility to identify and inform the proper authorities of unsafe work.

Audio and video recording

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan.

Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study and is not to be used or distributed for any other purpose without prior written consent from the content author(s).

Only those items specifically authorized by the instructor may be brought into the exam facility. Students must not bring any unauthorized electronic device into an examination room, including cell phones or other devices.

Student services and support

Health & Wellness Support

Counselling and Clinical Services

Free, short-term, appointment-based counselling and psychiatric services. Also offers drop-in workshops. Book an initial consultation. Visit uab.ca/CCS to learn more.

Wellness Supports Social Workers

Free one-on-one support for students in the areas of housing, finances, academics, personal wellness, life skill development, family dynamics, system navigation, and any area of life where there is a desire to invite change. Visit uab.ca/wellness to learn more.

Sexual Assault Centre

Free, anonymous, and confidential drop-in counselling. Visit uab.ca/UASAC to learn more.

The Office of Safe Disclosure & Human Rights (OSDHR)

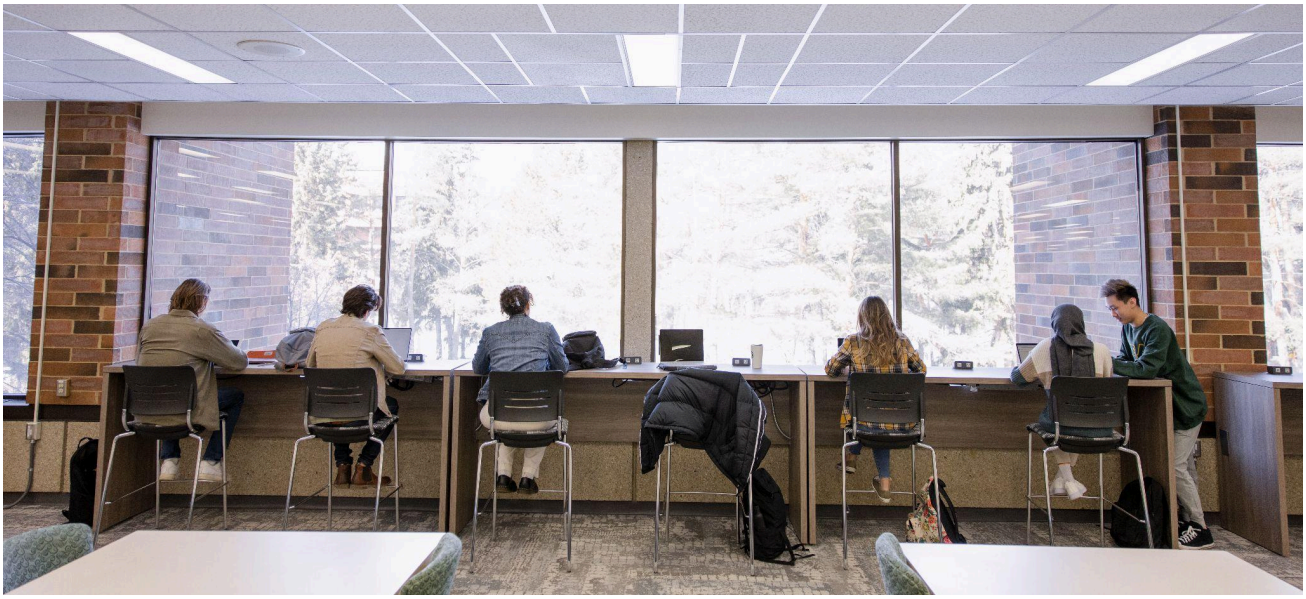
The OSDHR advises confidentially on sensitive issues you may not feel comfortable solving on your own. Contact the OSDHR if you want to get help or to make a report while keeping your privacy. Visit uab.ca/OSDHR to learn more.

HIAR (Helping Individuals at Risk)

If you're worried about someone, contact HIAR, who can help assess risk and connect individuals to support. Learn more at uab.ca/HIAR.

Immediate External Supports

Health Link Alberta: 811
Suicide Crisis Helpline: 988



Academic support

Academic Success Centre

Access to a variety of services to maximize your academic success. Learn more at uab.ca/ASC.

Accessibility Resources

Connects students with disabilities to accommodations. Learn more at uab.ca/Access under accommodations + accessibility.

Decima Robinson Support Centre

Academic support for 100- or 200-level introductory calculus, linear algebra and statistics courses. Visit uab.ca/DSC to learn more.

Engineering Student Success Centre

The Faculty of Engineering provides drop-in tutoring for first-year courses. Visit uab.ca/ESSC to learn more.

Office of the Student Ombuds

Call for complex problems and conflict mediation. Learn more at uab.ca/ombuds.

UNIVERSITY OF ALBERTA

Financial support

Student Service Centre

For awards and other funding support. Learn more at uab.ca/ask.

Campus Food Bank

The Campus Food Bank Society is an independent charity supporting University of Alberta students, faculty, staff, and alumni for up to five years. For additional information visit their website at campusfoodbank.com.

ECE 447: Data Analysis and Machine Learning for Engineers
Winter 2026

1 Learning Outcomes (continued)

The final letter grade will be assigned in the end of the term to align with a student's performance and level of achievement for this course. For the reference only, an example for the process of converting the term marks to letter grade is shown below:

	Approx. Range	Letter Grade	Grade Point Value
Excellent	**top 15%** e.g., 85–100%	A+	4.0
		A	4.0
		A–	3.7
Good	**next 20%** e.g., 65–85%	B+	3.3
		B	3.0
		B–	2.7
Satisfactory	**next 15%** e.g., 50–65%	C+	2.3
		C	2.0
		C–	1.7
Minimal Pass	**next 10%** e.g., 40–50%	D+	1.3
		D	1.0
Failure	**rest** e.g., 0–40%	F	0.0

Note that those ranges are not fixed and will be adjusted to reflect students' achievement performance in a class.

2 Assignment/Project

Assignments will be posted on Mondays (9:30 AM); they will be due in 2 weeks at 9:30 AM, also on Mondays. The coding component of the homework are expected to be in Jupyter notebooks format (code with comments). It is assumed the assignments will be done individually.

There will be one project. It will be done in groups of 4. The final deadline for the project report is Monday, April 20, 11:59 PM. Student will be delivering a project presentation toward the end of the course (see Syllabus).

For assignments and project deadlines, there will be a non-negotiable 15-minute grace period. Any assignment or project submitted after this grace period will receive a grade of zero.

3 Midterm Exam

Students who do not take the midterm exam at the scheduled date and time will receive a grade of zero. Exceptions will be granted only in cases of formally documented and university-approved absences (e.g., official medical documentation or other recognized accommodations submitted through the appropriate university channels).

Personal emails sent after the exam stating illness or other reasons, without prior approval and proper documentation, will not be considered. Students are responsible for understanding and adhering to this policy. Please plan accordingly.

4 Anaconda Jupyter Notebook/CoLab

All students are required to use only one, platform independent, Anaconda environment. You are encouraged to install the software on your home machine or laptop so you can work on lab problems at your leisure. The software is available at: `https://www.anaconda.com`.

You can also use Google CoLab – `https://www.youtube.com/watch?v=inN8seMm7UI`.

5 Canvas

You will use Canvas throughout the course to access course information, lecture notes, seminar information, sample code, your marks, announcements, and other course-related material. You are expected to keep current and read the Announcements on a regular basis. Lecture notes, which are the copyright of the University of Alberta, will be posted under Course Content. You may access the course at `https://canvas.ualberta.ca` using your CCID and password. Important legal note: Your activity on Canvas is logged and can be tracked.

6 Course Notes and Syllabus

They will be available via Canvas.

7 Recording of Lectures

Recording is permitted ONLY with the prior written consent of the professor or if recording is part of an approved accommodation plan.

8 Instructional Material

Recommended Text:

John D. Kelleher, Brian Mac Namee and Aoife D'Arcy,
Fundamentals of Machine Learning for Predictive Data Analytics
MIT Press; 2nd Edition (October 20, 2020).

Additional Text:

Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong,
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Emmanuel Raj,
Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale
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